

Input / output

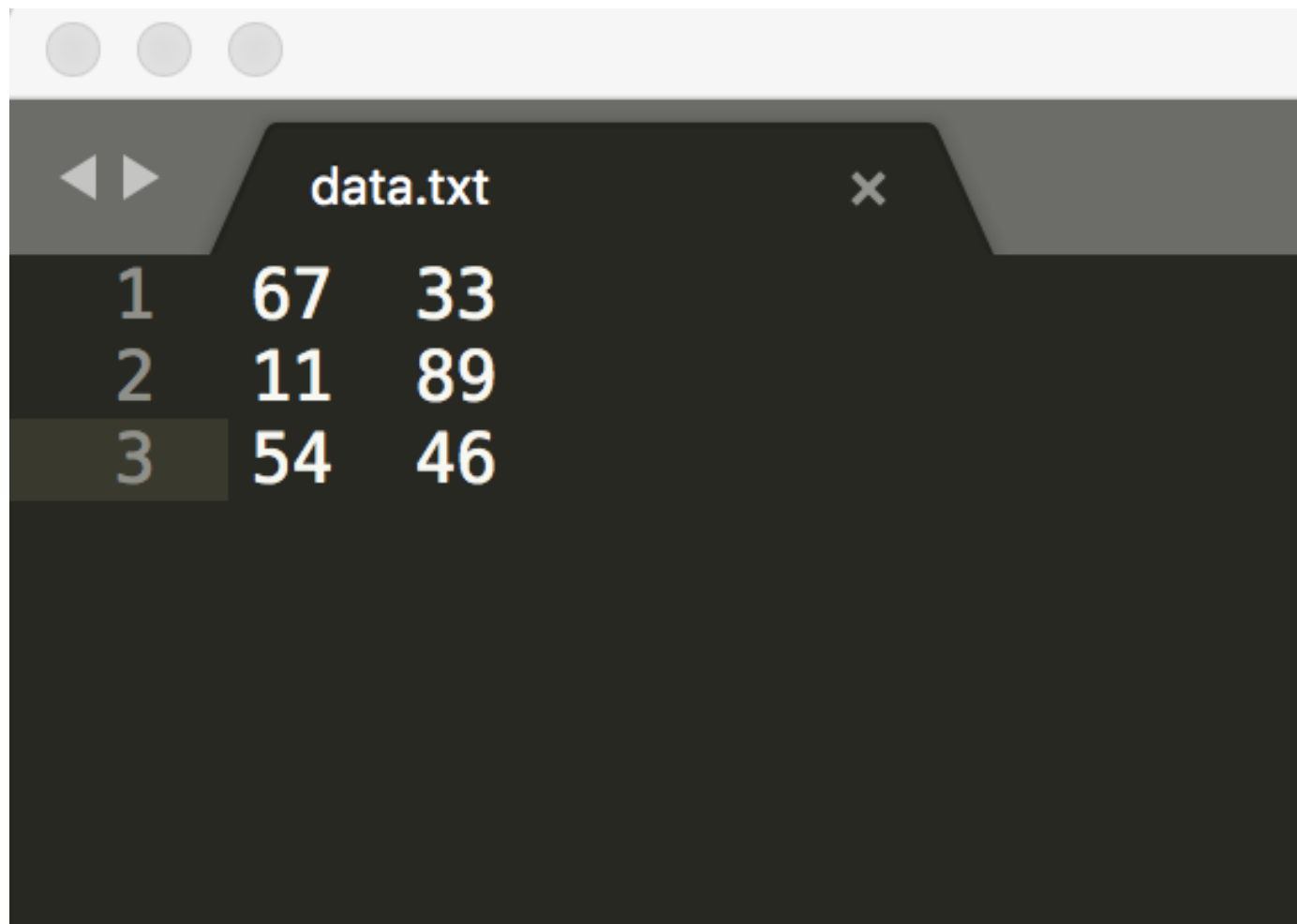
- Input
 - Read a files from your handdisk
 - Command line arguments
 - Interactive input
- Output
 - Terminal output (print)
 - Write files to your harddisk

Input / output

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 - Read a files from your handdisk
 - Command line arguments
 - ~~Interactive input~~ *for now*
- Output
 - Terminal output (print)
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Make a data file

Open a new file in your code editor. Write the following and save it as **data.txt**.

A screenshot of a code editor window. The window has a title bar with three window control buttons (minimize, maximize, close) on the left. Below the title bar is a tab bar with a single tab labeled 'data.txt' and a close button (X) on the right. The editor area has a dark background and contains three lines of text. The first line is '1 67 33', the second line is '2 11 89', and the third line is '3 54 46'. The third line is highlighted with a light gray background. The text is in a monospaced font.

```
1 67 33
2 11 89
3 54 46
```

Read a file

```
input_file_name = 'data.txt'  
data_file = open(input_file_name, 'r')
```

```
file_content = data_file.read()  
print(file_content)
```

```
data_file.close()
```



Open a file

```
data_file = open( 'data.txt', 'r' )
```

Read file line by line

```
input_file_name = 'data.txt'
data_file = open(input_file_name, 'r')

for line in data_file:
    print(line)
```

Exercise

Make this code also print the sum of the two numbers on each line:

```
input_file_name = 'data.txt'
data_file = open(input_file_name, 'r')

for line in data_file:
    # your code here...
```

Look up the
split method!
(pydoc str.split)

Solution?

```
input_file_name = 'data.txt'
data_file = open(input_file_name, 'r')

for line in data_file:
    fields = line.split()
    n1 = fields[0]
    n2 = fields[1]
    print(n1 + n2)
```

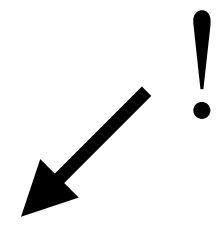
Solution!

```
input_file_name = 'data.txt'
data_file = open(input_file_name, 'r')

for line in data_file:
    fields = line.split()
    n1 = int(fields[0])
    n2 = int(fields[1])
    print(n1 + n2)
```

Write to a file

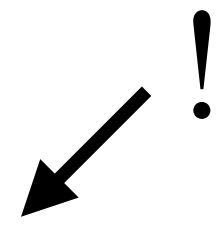
```
output_file_name = 'results.txt'  
out_file = open(output_file_name, 'w')
```



```
print('First line', file=out_file)  
print('Second line', file=out_file)
```

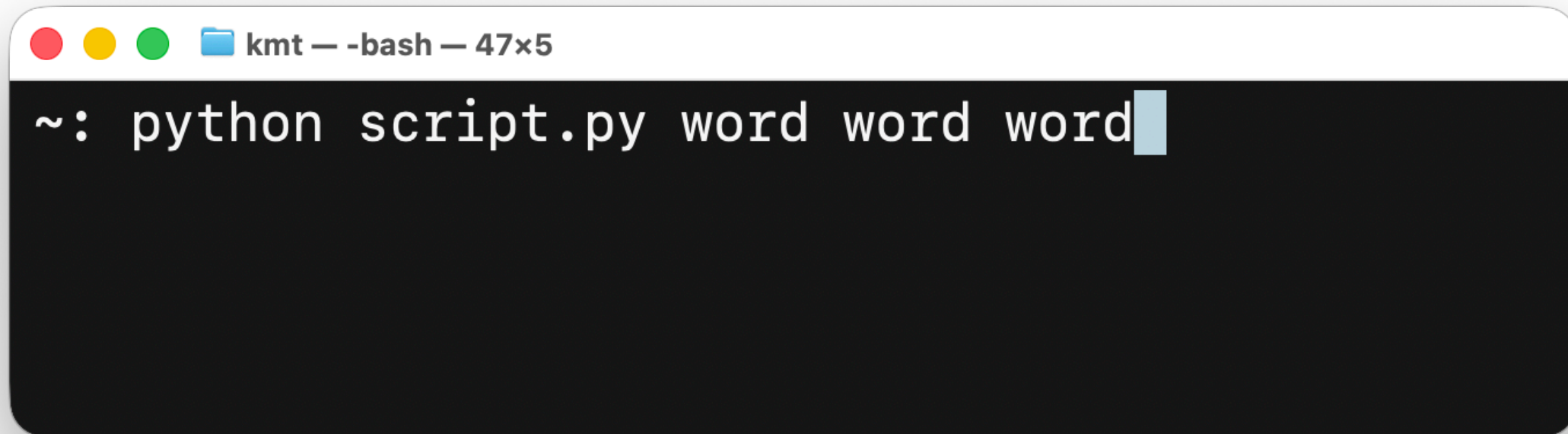
Exercise

```
output_file_name = 'results.txt'  
out_file = open(output_file_name, 'w')  
  
print('First line', file=out_file)  
print('Second line', file=out_file)
```



Udbyg jeres script så i skriver summen af de to tal i hver linje til en ny fil: **results.txt**

Command line arguments

A terminal window with a title bar containing three colored circles (red, yellow, green) and a folder icon, followed by the text "kmt — -bash — 47x5". The terminal content shows a prompt "~:" followed by the command "python script.py word word word" and a light blue cursor at the end of the line.

```
kmt — -bash — 47x5
~: python script.py word word word
```

```
# script.py:
```

```
from sys import argv
```

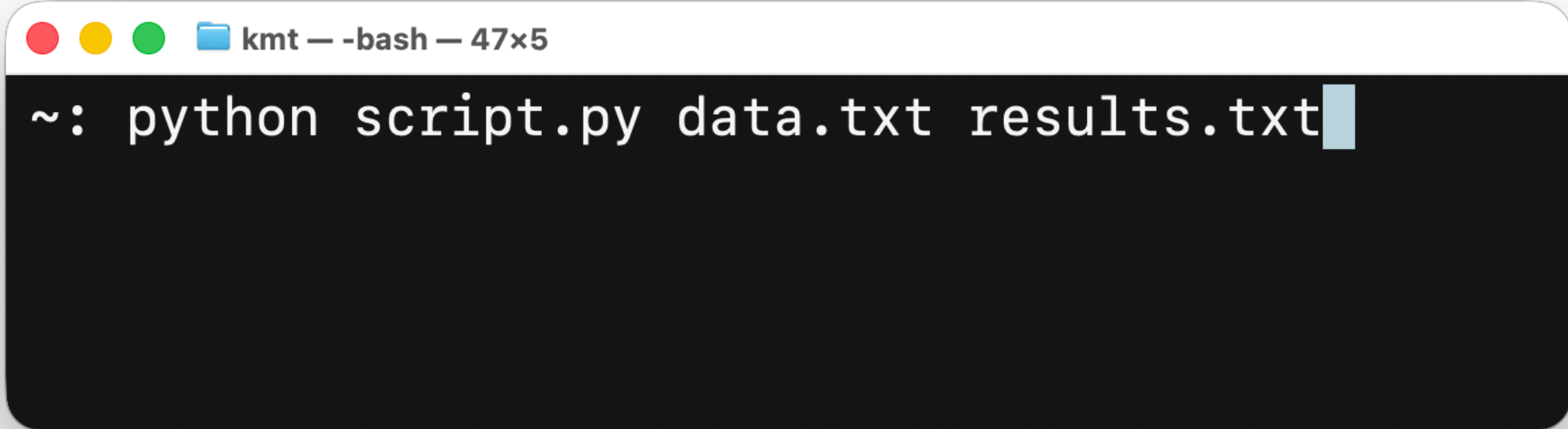
```
print(argv)
```

Exercise

Udbyg dit script så input og output fil gives som argumenter til scriptet

```
from sys import argv
```

```
input_file = open(argv[1], 'r')  
output_file = open(argv[2], 'w')
```

A terminal window with a title bar containing three colored circles (red, yellow, green) and a folder icon. The title text is "kmt — -bash — 47x5". The terminal content shows a prompt "~:" followed by the command "python script.py data.txt results.txt" and a light blue cursor at the end of the command.

kmt — -bash — 47x5

```
~: python script.py data.txt results.txt
```